

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0150

Peach Bottom 500 kV
September 19, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0150. The Interconnection Customer proposes a 576.6 MW natural gas combined cycle facility interconnecting at the Peach Bottom 500 kV substation in the Pennsylvania area. The request was received on August 23, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 94 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$599,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

CURECANTI SUBSTATION / MOSER GENERATING STA 345.0 kV: 101.0% loading (1,413 MVA) following the outage of the TOMBSTONE / BXBY HILL SUBSTATION 500.0 kV; DFAX = 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CURECANTI SUBSTATION / MOSER GENERATING STA 345.0 kV	Reconductor line	\$8,429,000	7% (\$599,000)
	Total allocated to project		\$599,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.